Quality Lab Manual

Operating System Lab
2ICPC204

Experiment No. 1

Installation of Linux/UNIX operating System.

Course Instructor -
PROF.MADHURI J SIDDHARAPU
ASSISTANT PROFESSOR

Experiment No. 1

Title of Experiment: Installation of Linux/UNIX operating system

Aim of Experiment: To Install Linux/UNIX operating System on your Computer System and configure basic system settings.

System Requirements –

1. **Hardware Requirements:**
    - Processor: 1 GHz or faster, 64-bit or 32-bit processor
    - RAM: Minimum 2 GB (4 GB or more recommended)
    - Hard Disk: At least 25 GB of free space
    - USB Drive/DVD: Bootable installation medium (at least 8 GB for USB or a DVD drive)
2. **Software Requirements:**
    - Operating System: Linux/UNIX distribution (e.g., Ubuntu, Fedora, CentOS, Debian)
    - Bootable Software: Tools like Rufus (for creating a bootable USB) or a bootable DVD
    - Installation Media: ISO file of the Linux/UNIX distribution

Experiment Outcomes –

1. To Understand how to install a Linux/UNIX operating system from bootable media such as a USB or DVD..
2. To Configure basic system settings like language, time zone, partitioning, and user accounts.
3. To Understand the concept of disk partitioning (e.g., root, swap, home partitions) and how to allocate disk space
4. To Understand the role of bootloaders (e.g., GRUB) and how to configure it during the installation process.
5. To learn to install Linux alongside another operating system (dual boot) or within a virtualized environment.

Description -

1) What is a Linux distribution?

- A Linux distribution, or distro, is an installable operating system built from the Linux kernel, supporting user programs, repositories, and libraries. Each vendor or community's version is a distro.
- Because the Linux operating system is open-sourced and released under the GNU General Public License (GPL), anyone can run, study, modify, and redistribute the source code, or even sell copies of their modified code.

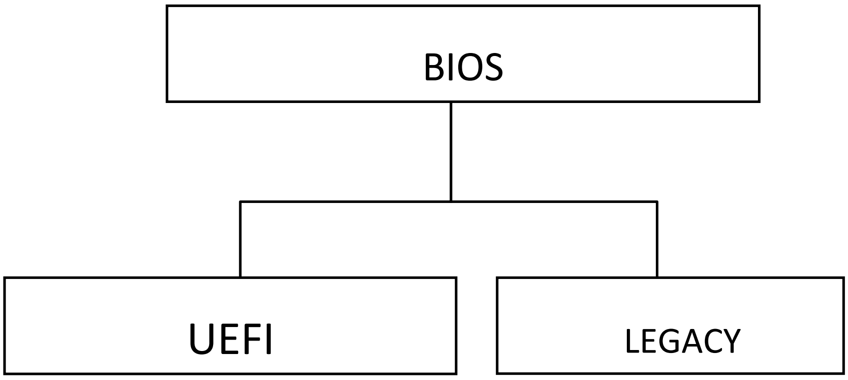
- Popular Linux distros are:

- Android
- Arch Linux
- Centos
- Debian
- Elementary OS
- Fedora Linux
- Gentoo Linux
- Kali Linux
- Linux Lite
- Linux Mint
- Manjaro Linux
- MX Linux
- openSUSE
- Pop!_OS
- Puppy Linux
- Slackware
- Solus
- SUSE
- Ubuntu and all its versions (Gnome, Kubuntu –using KDE Plasma Desktop, Ubuntu mate, Xubuntu, and Lubuntu–just to name a few)
- Zorin OS

2) What is BIOS?

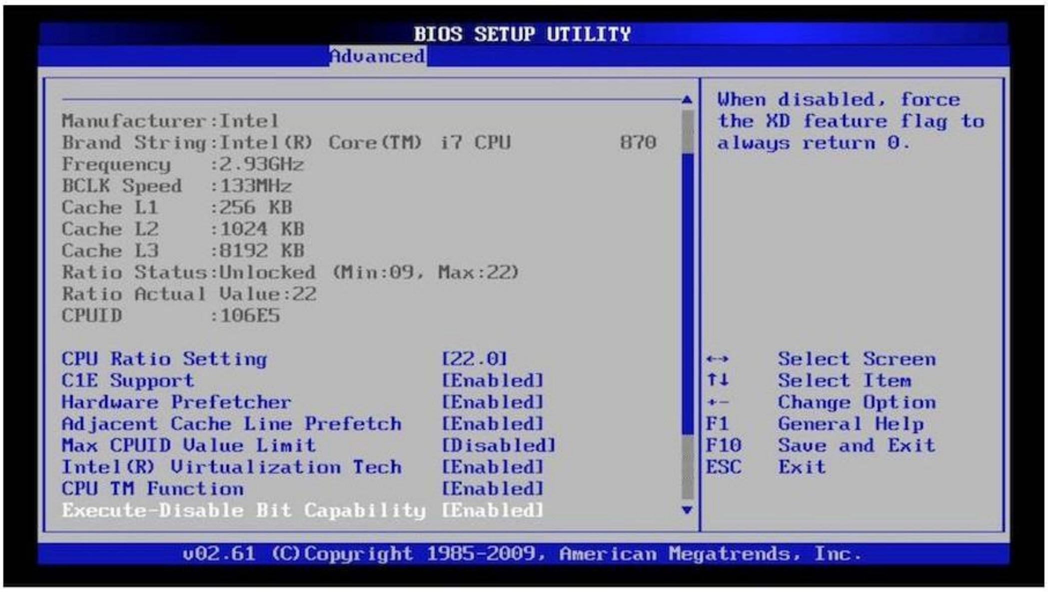
→ BIOS, pronounced "BYE-oss," stands for Basic Input Output System andis software stored on a small memory chip in your system's motherboard.When you boot up your system and look at the screen at the right time, you may see a start-up message that uses the term "BIOS."

BIOS is the first software that runs when you power on your system, performing an initial pack of diagnostic tests (POST, or Power On Self-Test) to see if there are any issues with the hardware. POST is the first step in your hardware's boot sequence. The machine won't continue withthe boot sequence if the POST fails.

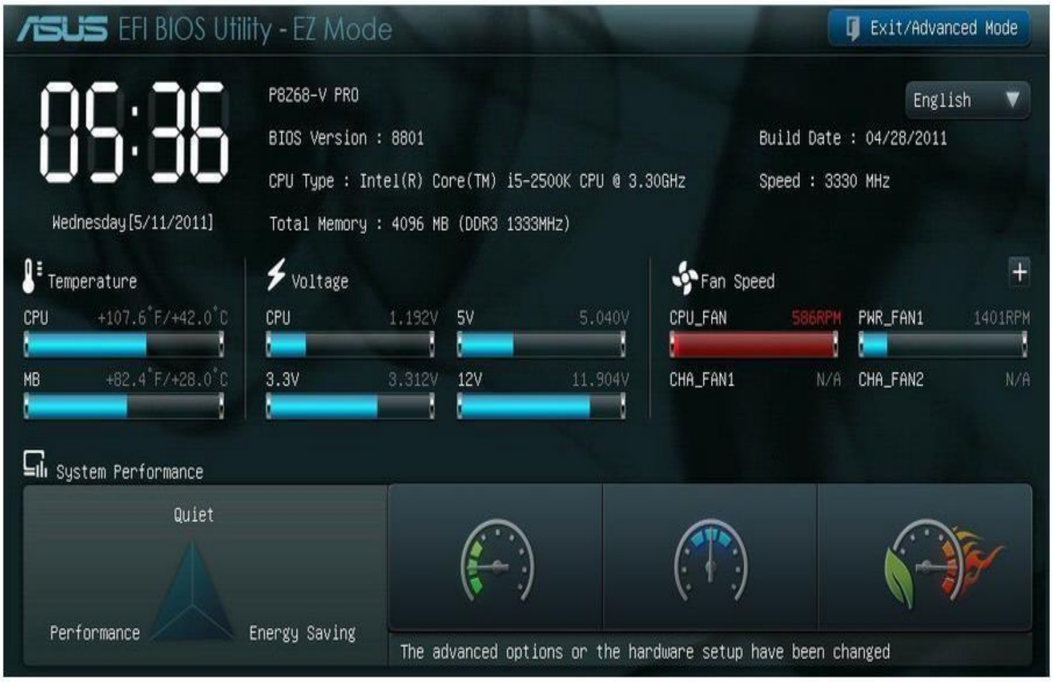


- **Legacy (1st generation) BIOS**: Older motherboards use legacy BIOS firmware to switch on the computer. Legacy BIOSes have additional constraints; even though similar to UEFI, they dictate how the CPU and components interact. These cannot recognize discs larger than 2.1 terabytes, and their installation systems have text-only menus.

- **UEFI: UEFI (Unified Extensible Firmware Interface)** may accept discs with a capacity of 2.2 TB or more by using the Master Boot Record (MBR) method as opposed to the more recent GUID Partition Table (GPT) technology. Even though Intel PCs are transitioning from traditional BIOS to UEFI firmware, Apple Mac PCs have never utilized BIOS.UEFI is the most recent among the two-boot software packages, introduced in 2002. UEFI has superior scalability, speed, programmability, and security compared to BIOS. UEFI does not need a separate bootloader application to run the OS. UEFI also features a superior user interface and higher performance generally.

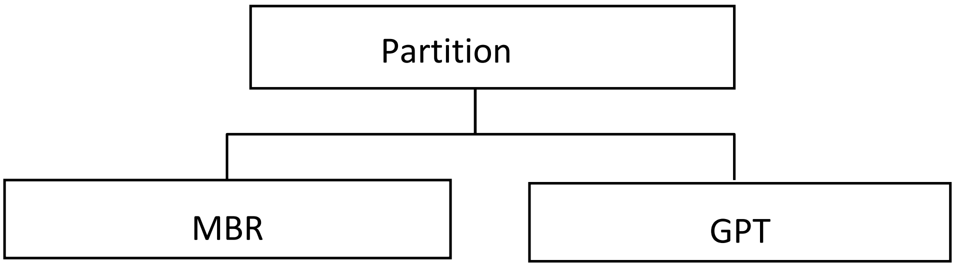


A Legacy BIOS configuration screen – Source



A UEFI BIOS Configuration screen – Source

3) There are two main types of Partition tables:



There are two main types of partition tables: MBR and GPT.

- MBR stands for Master Boot Record and is a bit of reservedspace at the beginning of the drive that contains informationabout how the partitions are organized. The MBR also contains code to launch the operating system, and it's sometimes called the Boot Loader.

- GPT is an abbreviation of GUID Partition Table, a newer standard slowly replacing MBR. Unlike an MBR partition table, GPT stores the data about how all the partitions are organized and how to boot the OS throughout the drive. Thatway if one partition is erased or corrupted, it's still possible to boot and recover some of the data.

- If you bought your computer within the last five years or so,it's very likely that it's using GPT partition tables rather thanthe older MBR tables.

- **What is ISO File?**

An ISO file is an exact copy of an entire optical disk such as a CD, DVD, or Blu-ray archived into a single file. This file, which is also sometimes referred toas an ISO image, is a smaller-sized duplicate of large sets of data.

- **Steps To Install Ubuntu on your System**

- [ ] Overview
- [ ] Download and install VirtualBox
- [ ] Download an Ubuntu Image
- [ ] Create a new virtual machine
- [ ] Define the Virtual Machine’s resources
- [ ] Insert the ISO File in the Virtual Which you have Created
- [ ] Installation Process

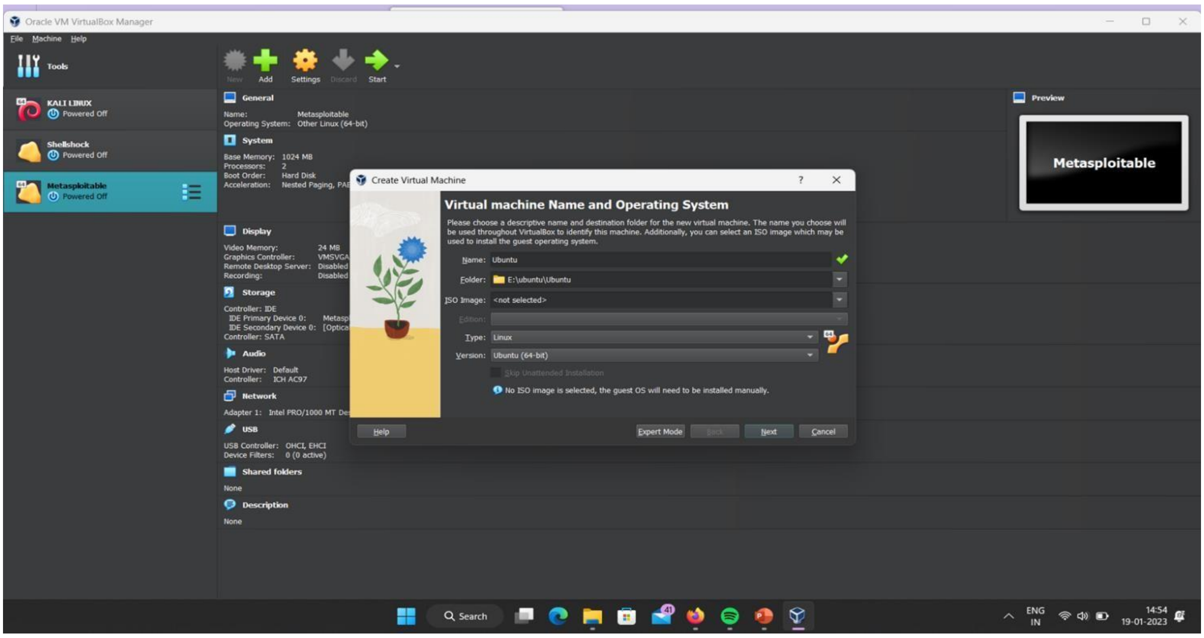
- ***Overview:***

In this tutorial, we’ll walk you through one of the easiest ways to try out UbuntuDesktop on a virtual machine. VirtualBox is a general-purpose virtualize that is available across Linux, Mac OS, and Windows. It’s a great way to experience Ubuntu regardless of your current operating system.
VirtualBox 7 and above includes a new feature called Unattended Guest OSInstall which significantly streamlines the setup experience for common operating systems like Ubuntu, making it easier than ever to get started.

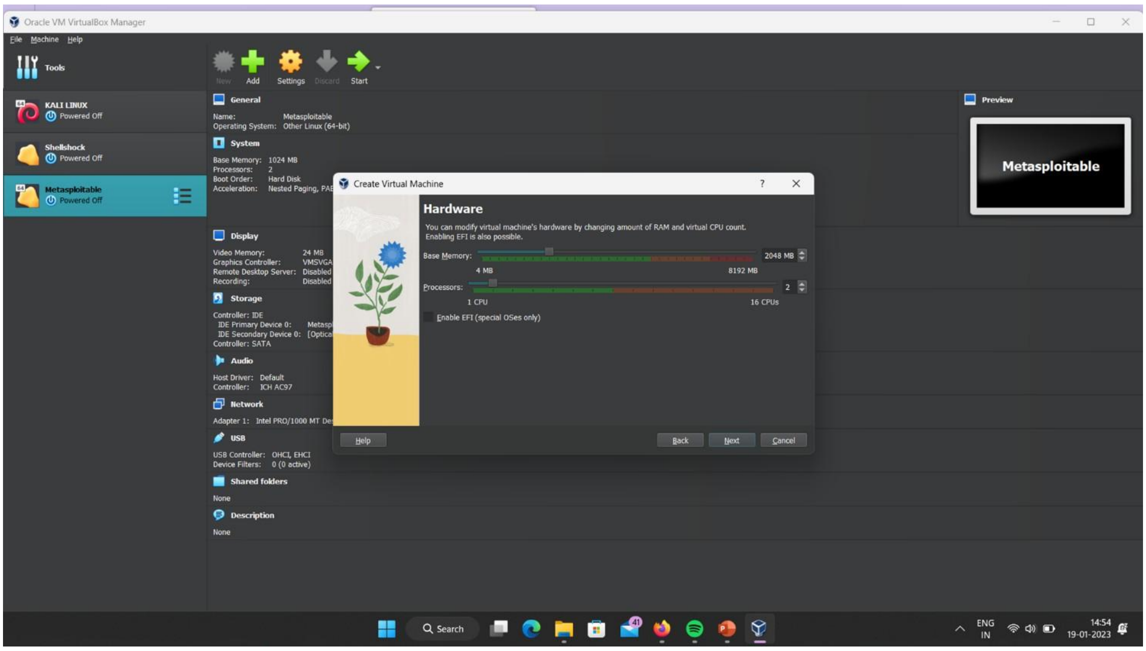
- ***Download and install VirtualBox:***
- You can download VirtualBox from the downloads page
- https://www.virtualbox.org/
- Once you have completed the installation, go ahead and run VirtualBox.

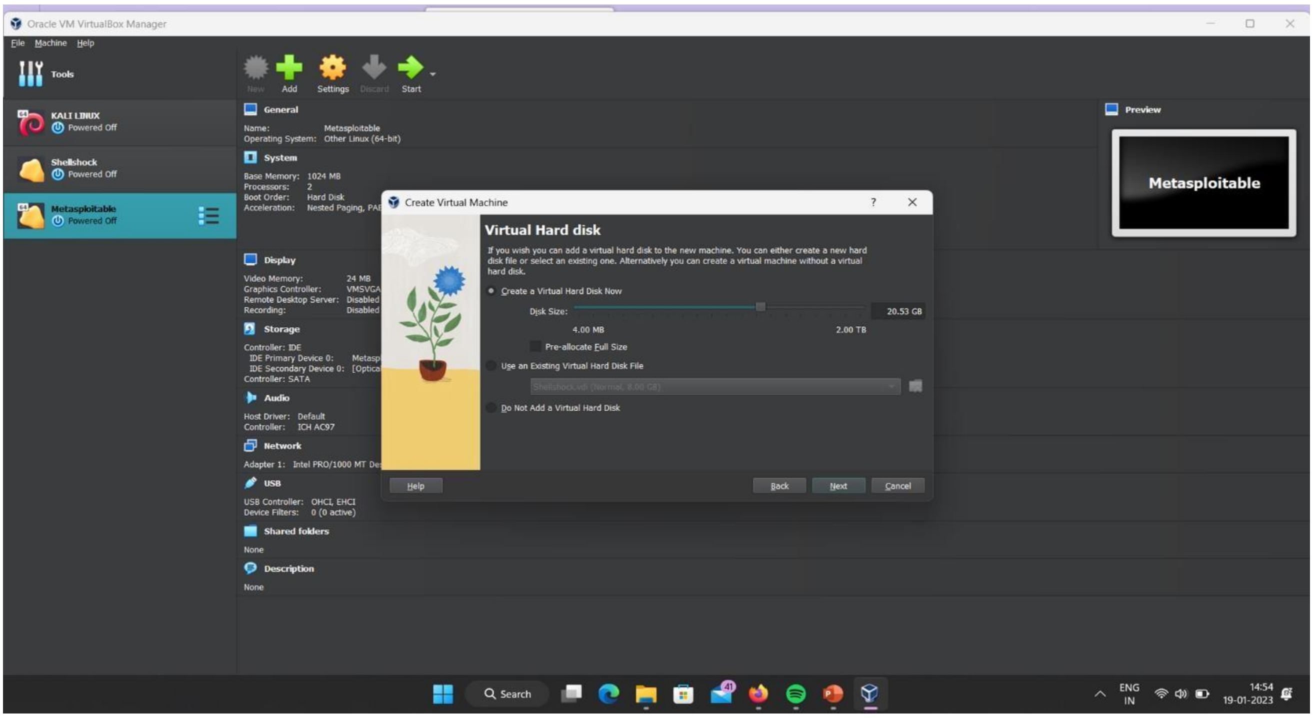
- ***Download an Ubuntu Image:***
  You can download an Ubuntu ISO File from the Official Website Of Ubuntu Make sure to save it to a memorable location on your PC! Forthis tutorial, we will use Ubuntu 22.04.1 LTS.

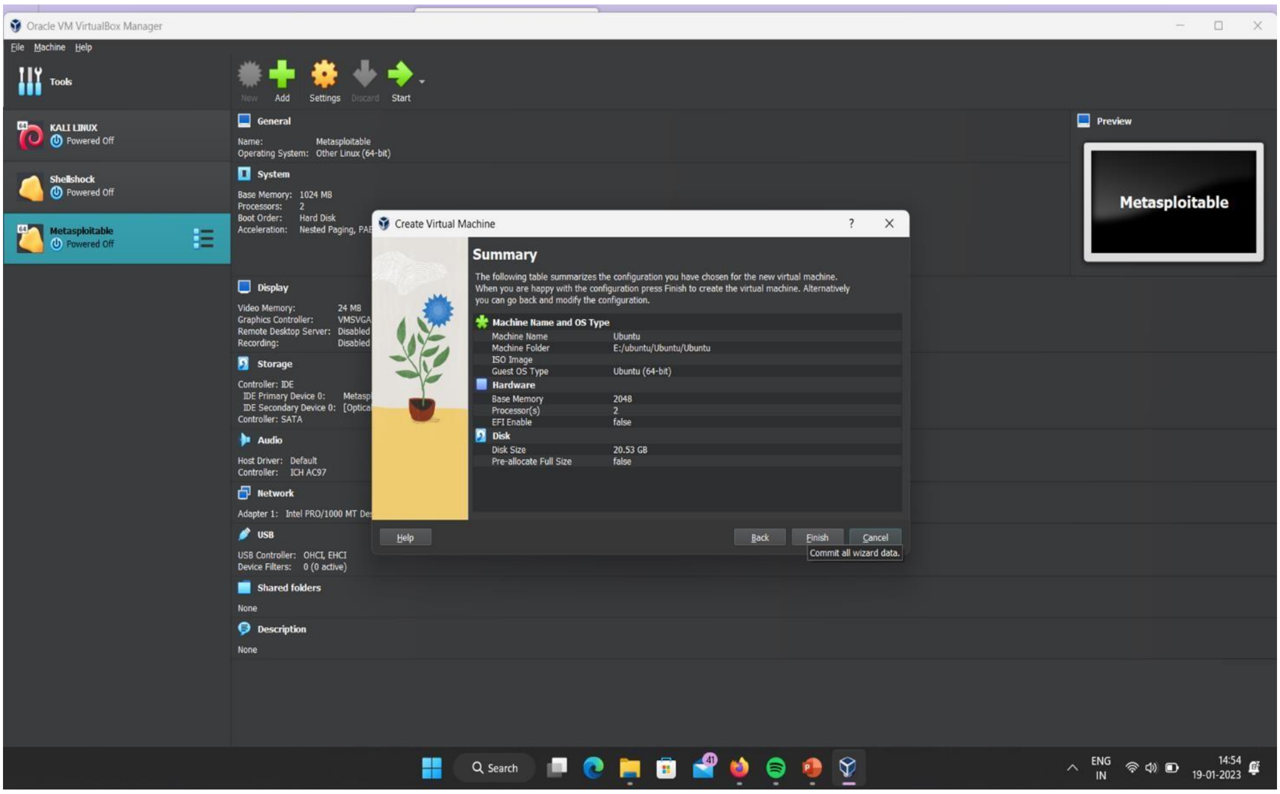
- **Create a new virtual machine:**



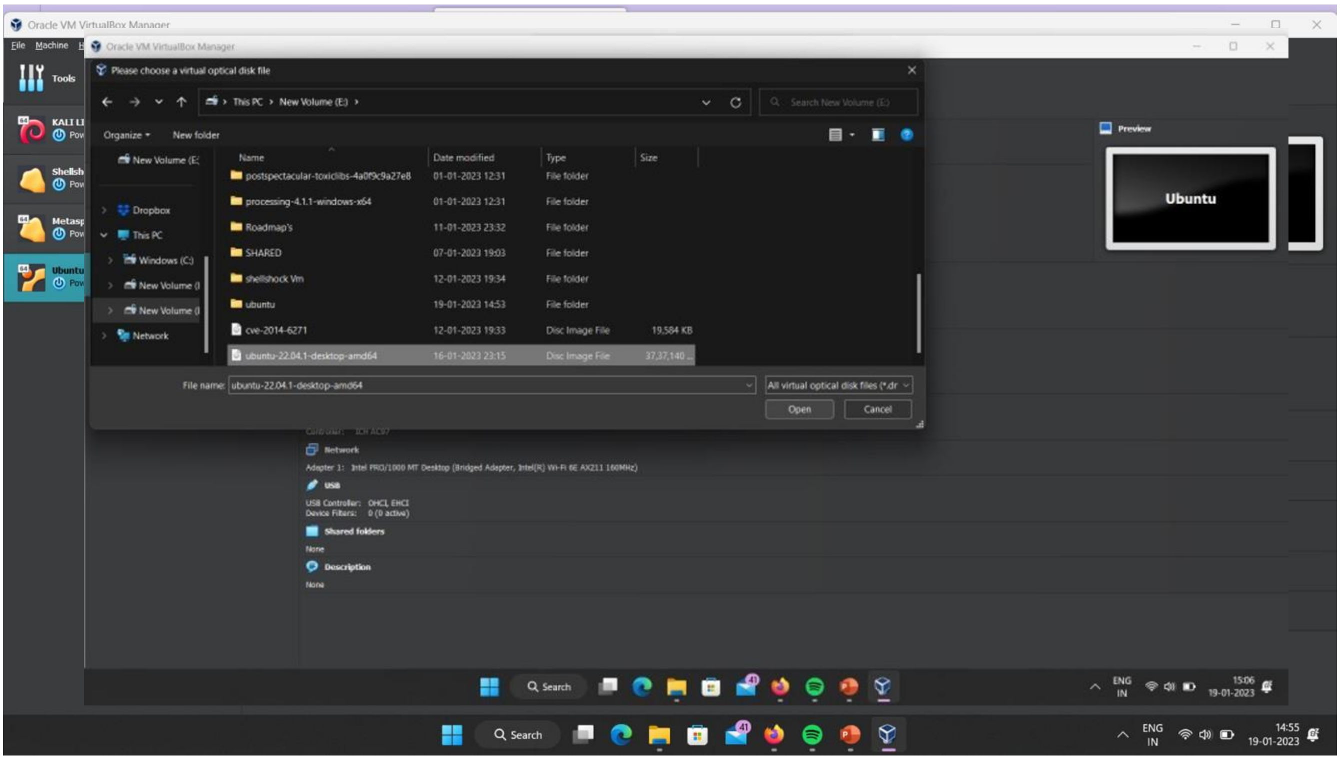
- **Define the Virtual Machine's resources:**



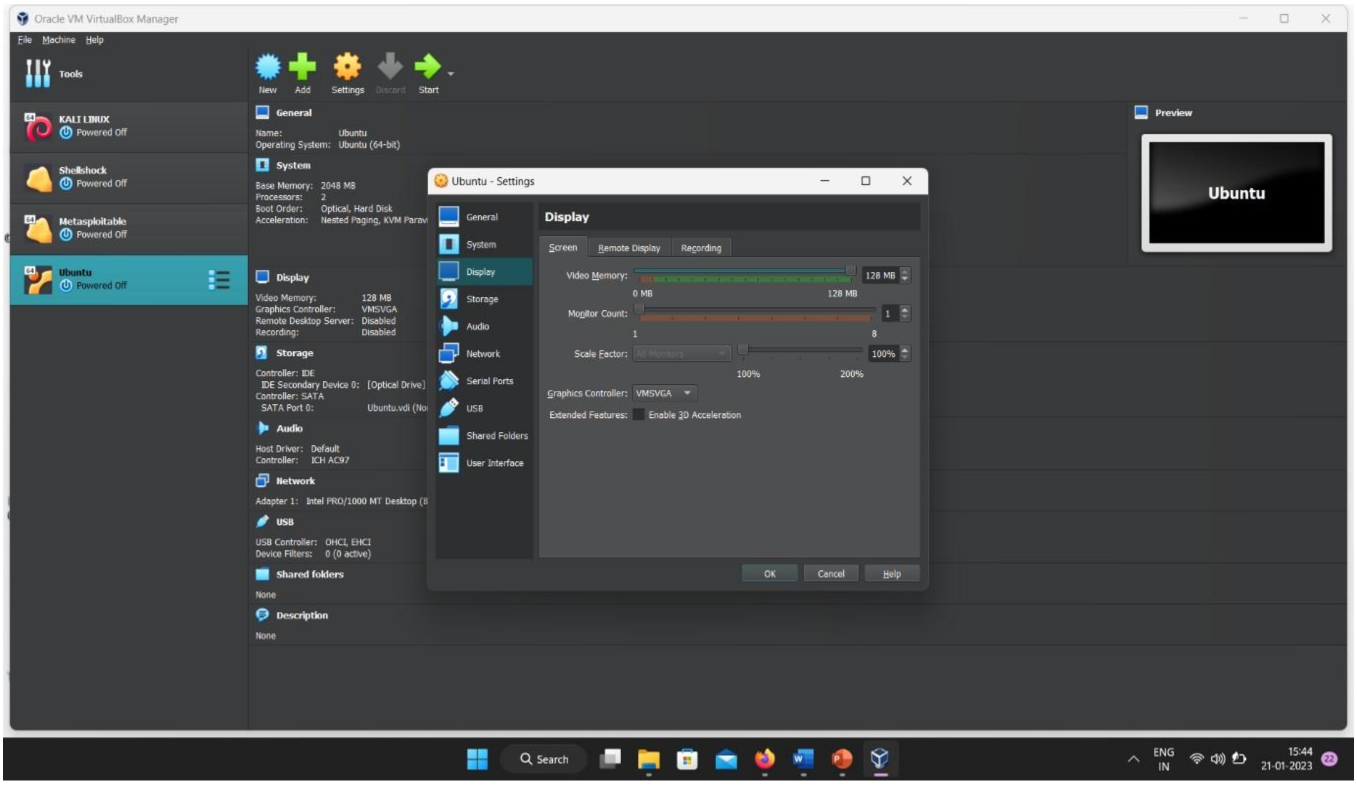

Oracle VM VirtualBox Manager
File Machine Help
Tools
New
Add
Settings
Discard
Start
KALI LINUX
Powered Off
Shellshock
Powered Off
Metasploitable
Powered Off
General
Name: Metasploitable
Operating System: Other Linux (64-bit)
System
Base Memory: 1024 MB
Processors: 2
Boot Order: Hard Disk
Display
Storage
Audio
Network
USB
Shared folders
Description
Preview
Metasploitable
Create Virtual Machine
Virtual Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.
Create a Virtual Hard Disk Now
Disk Size:
20.53 GB
4.00 MB
2.00 TB
Pre-allocate Full Size
Use an Existing Virtual Hard Disk File
Do Not Add a Virtual Hard Disk
Help
Back
Next
Cancel
Search
ENG
IN
14:54
19-01-2023


Oracle VM VirtualBox Manager
File Machine Help
Tools
KALI LINUX
Powered Off
Shellshock
Powered Off
Metasploitable
Powered Off
General
Name: Metasploitable
Operating System: Other Linux (64-bit)
System
Base Memory: 1024 MB
Processors: 2
Boot Order: Hard Disk
Preview
Metasploitable
Create Virtual Machine
Summary
The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.
Machine Name and OS Type
Machine Name Ubuntu
Machine Folder E:/ubuntu/Ubuntu/Ubuntu
ISO Image
Guest OS Type Ubuntu (64-bit)
Hardware
Base Memory 2048
Processor(s) 2
EFI Enable false
Disk
Disk Size 20.53 GB
Pre-allocate Full Size false
Help
Back
Finish
Cancel
Commit all wizard data.
Search
ENG
IN
14:54
19-01-2023

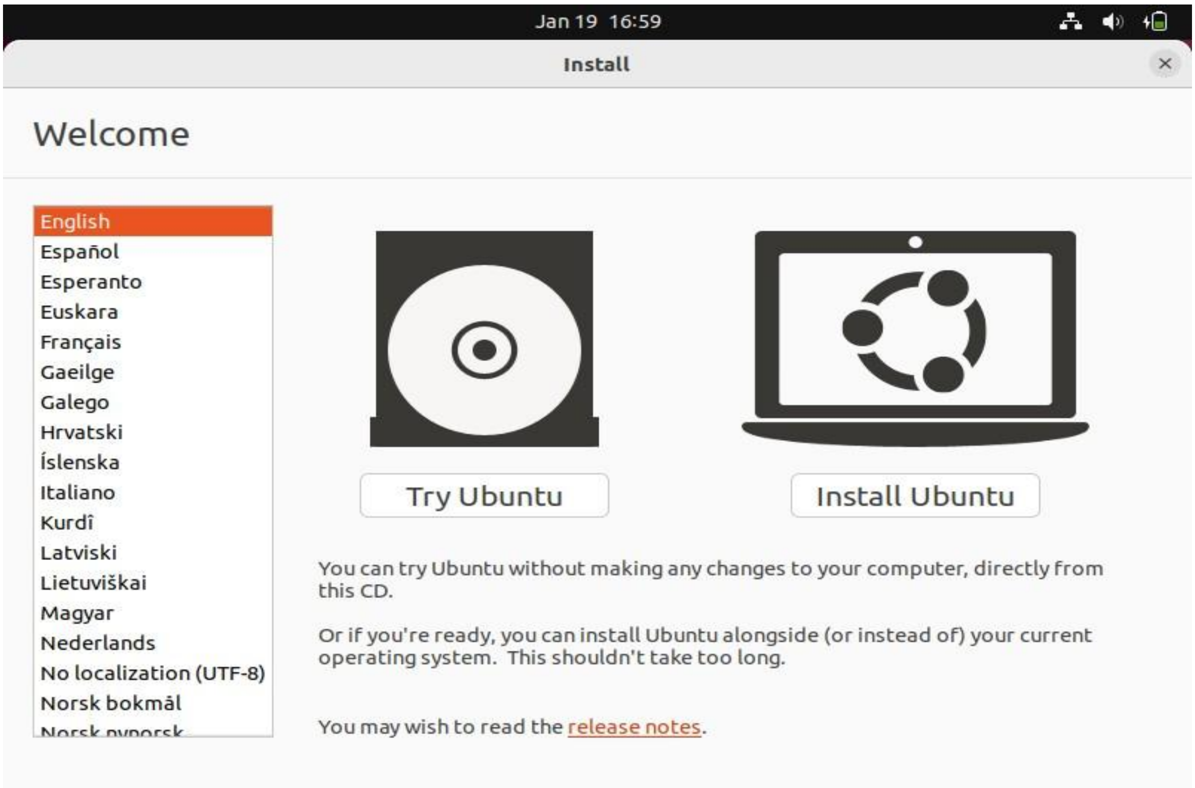
- **Configuring Network Settings in VirtualBox:**



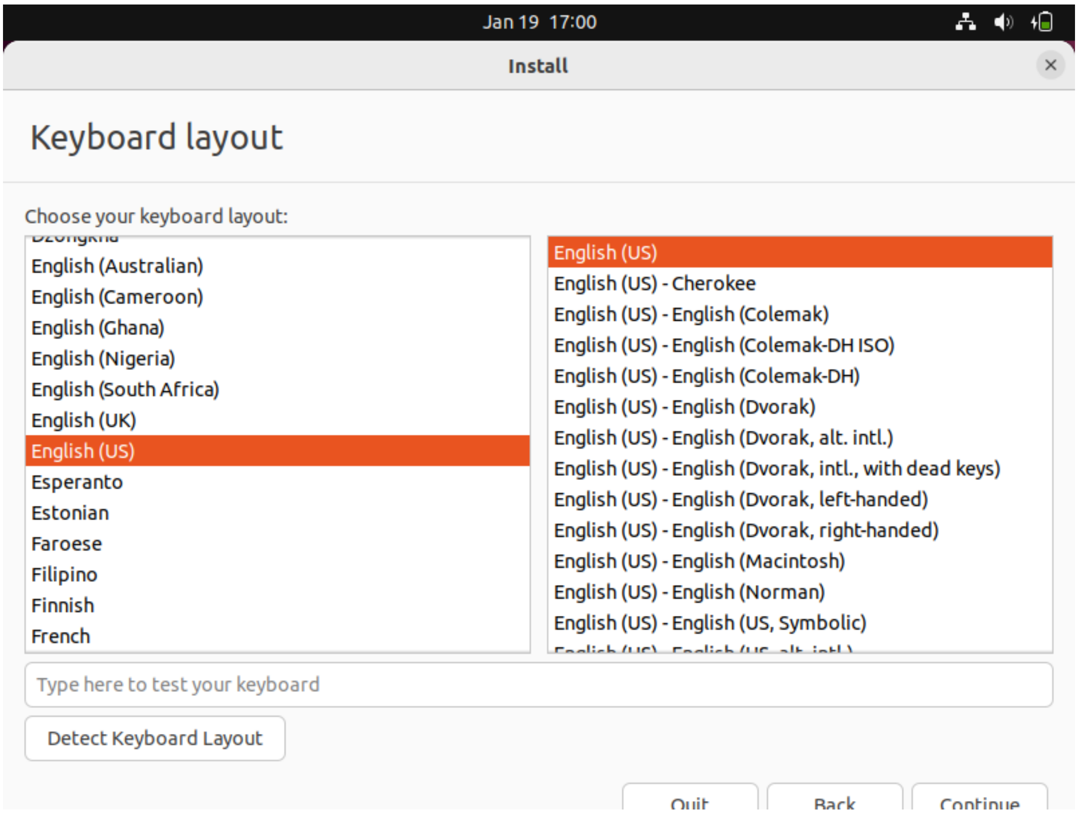
  - **Setting Up Graphics Settings:**
- **Insert the ISO File in the Virtual Which you have Created**



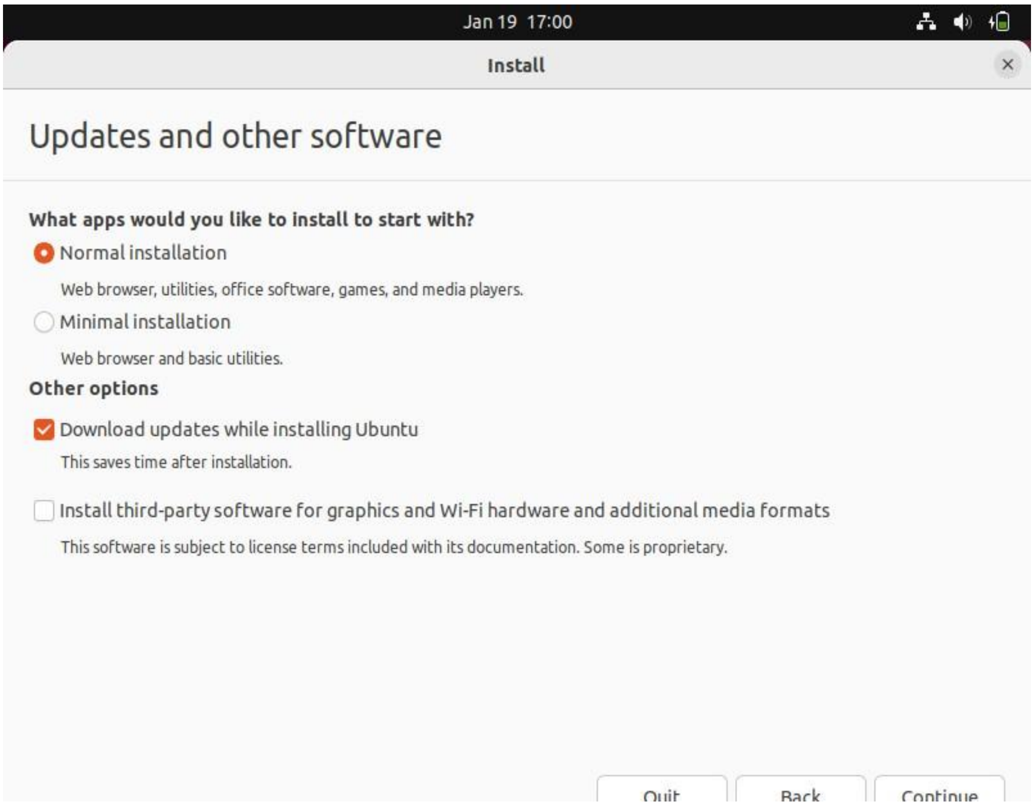
- **Installation Process:**
- *Select "Try or Install Ubuntu"*



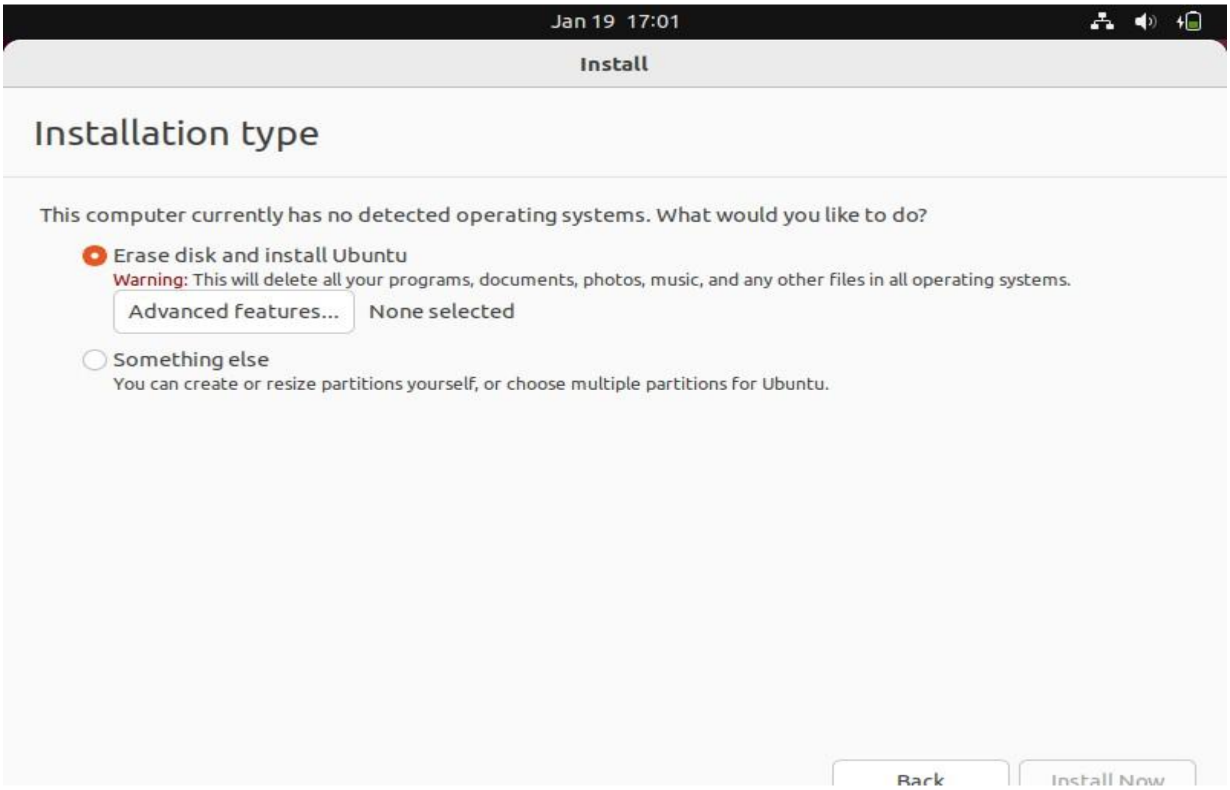
Click on *"Install Ubuntu".*



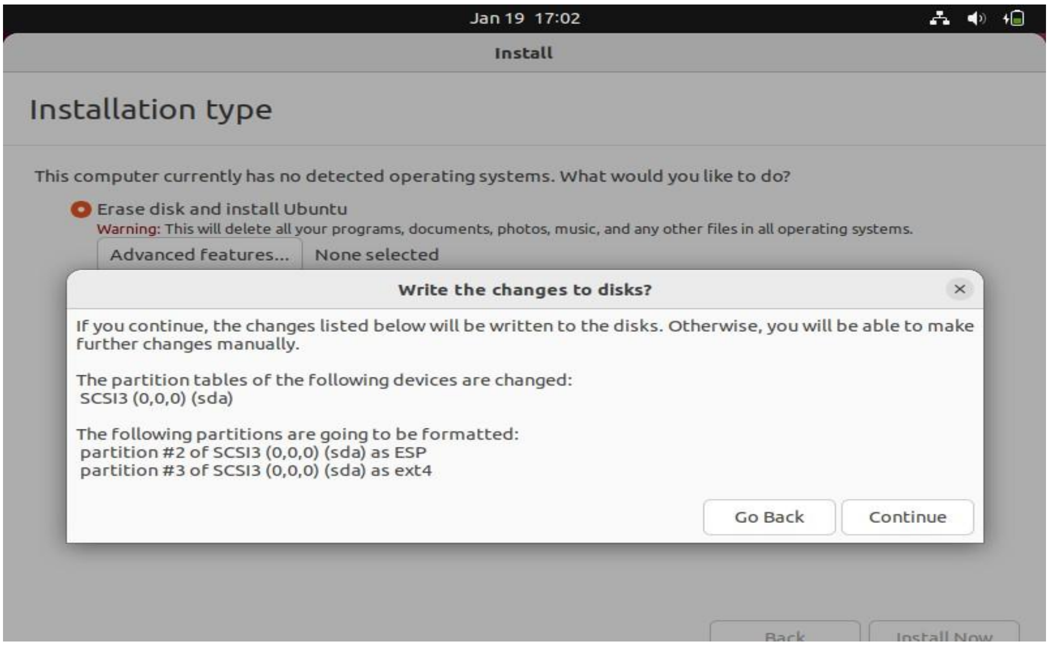
- Select your *"Keyboard Layout".*



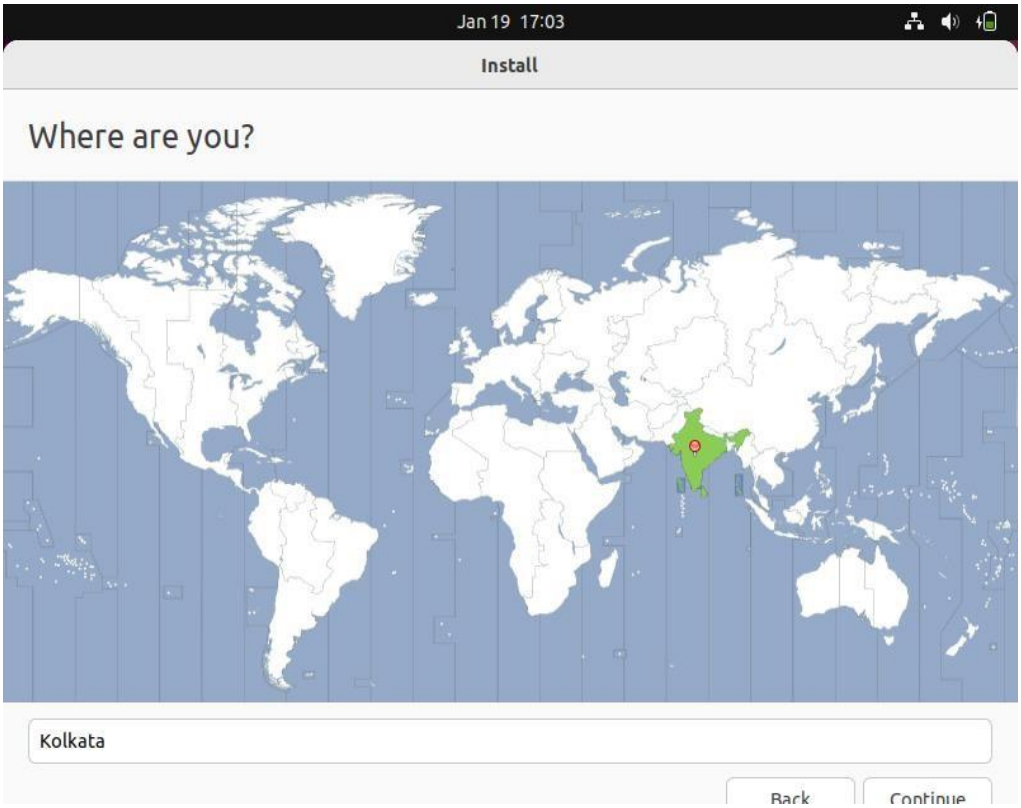
- Select Normal Installation If you have Good Specification and InternetConnection else Go with Minimal Installation.
- And You are Ready for the Next Step by Clicking Continue.



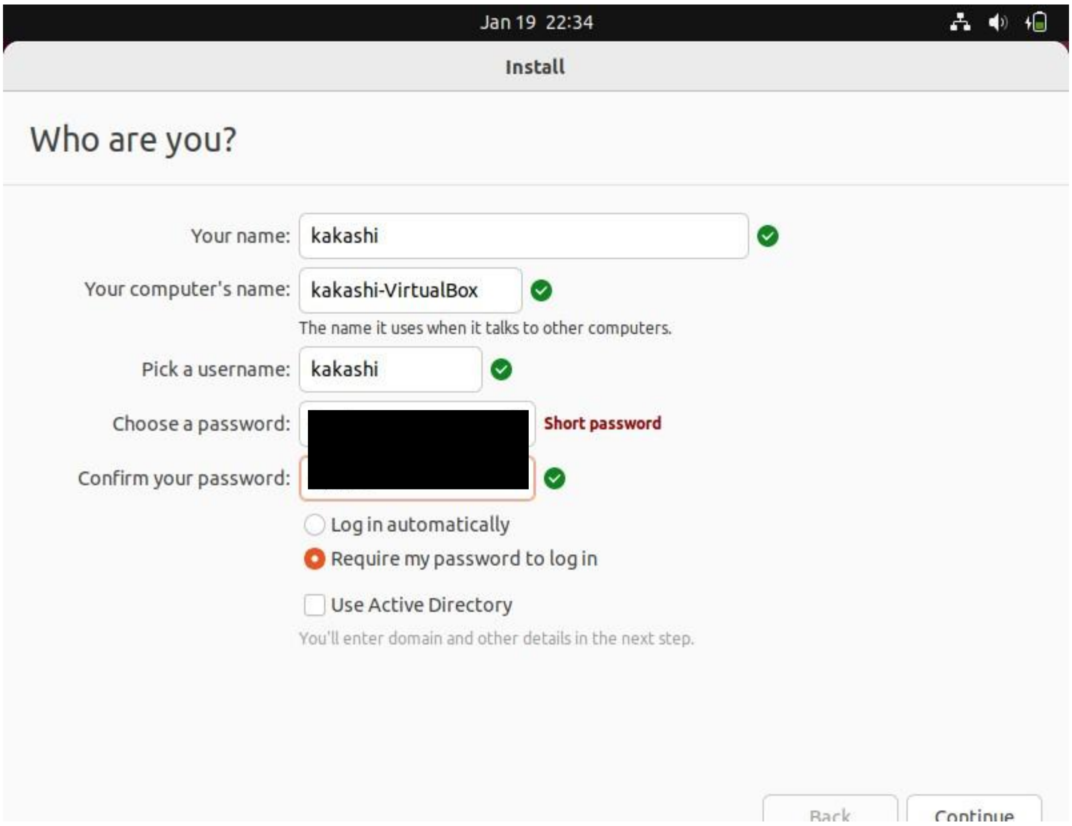
- Select On *"Erase Disk and Install Ubuntu"* or You can Create a partition(Primary and Secondary Partitions) and Install You Operating System.



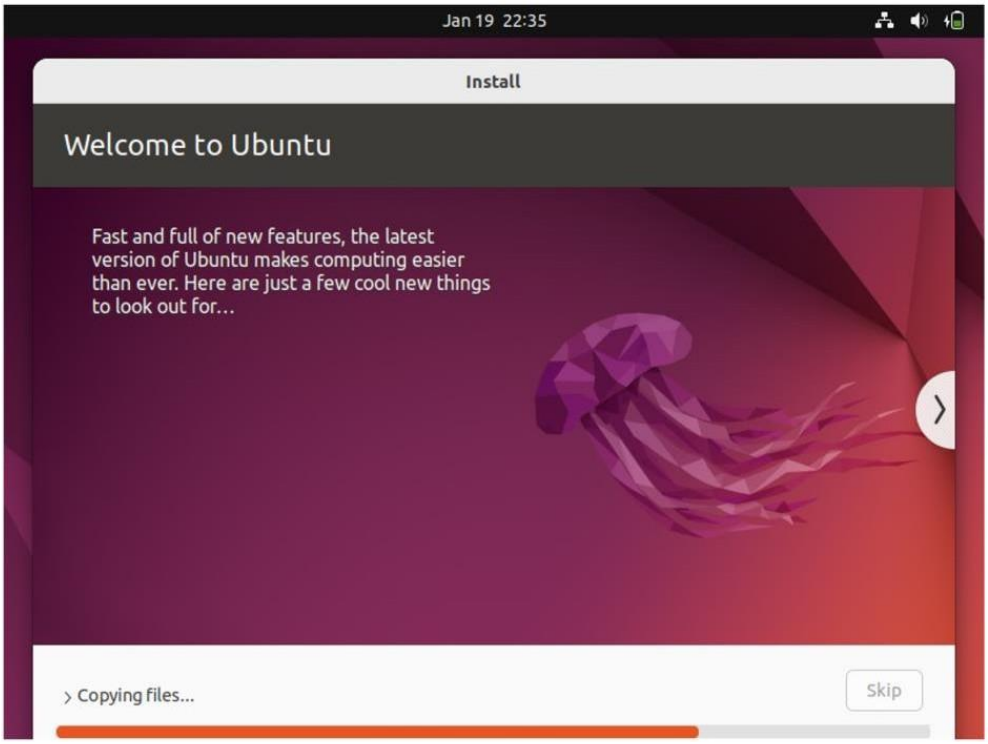
- Click On *"Continue"*.

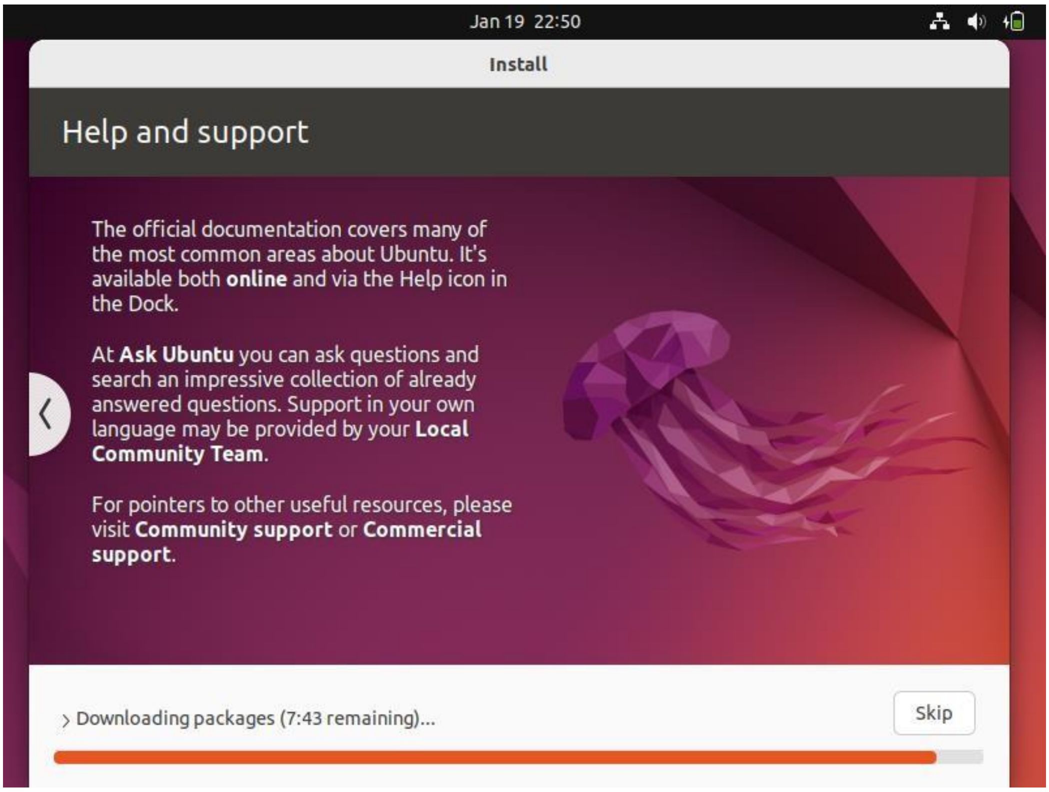


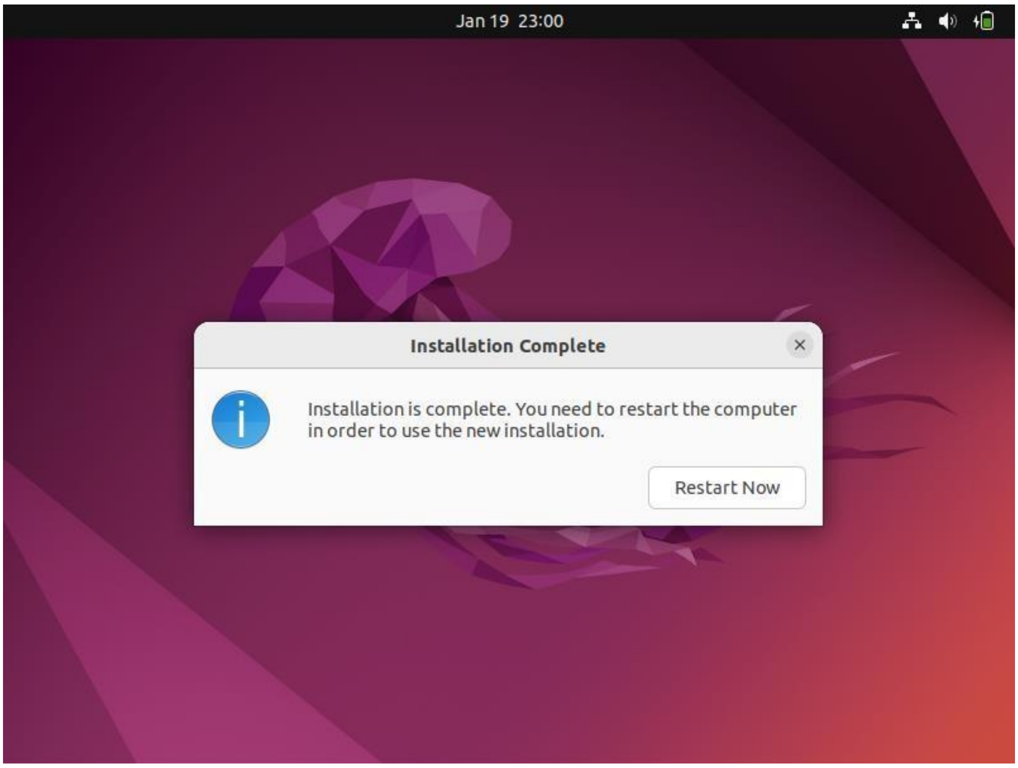
- Select your Location for Adujst Date and Time ,Click On Continue



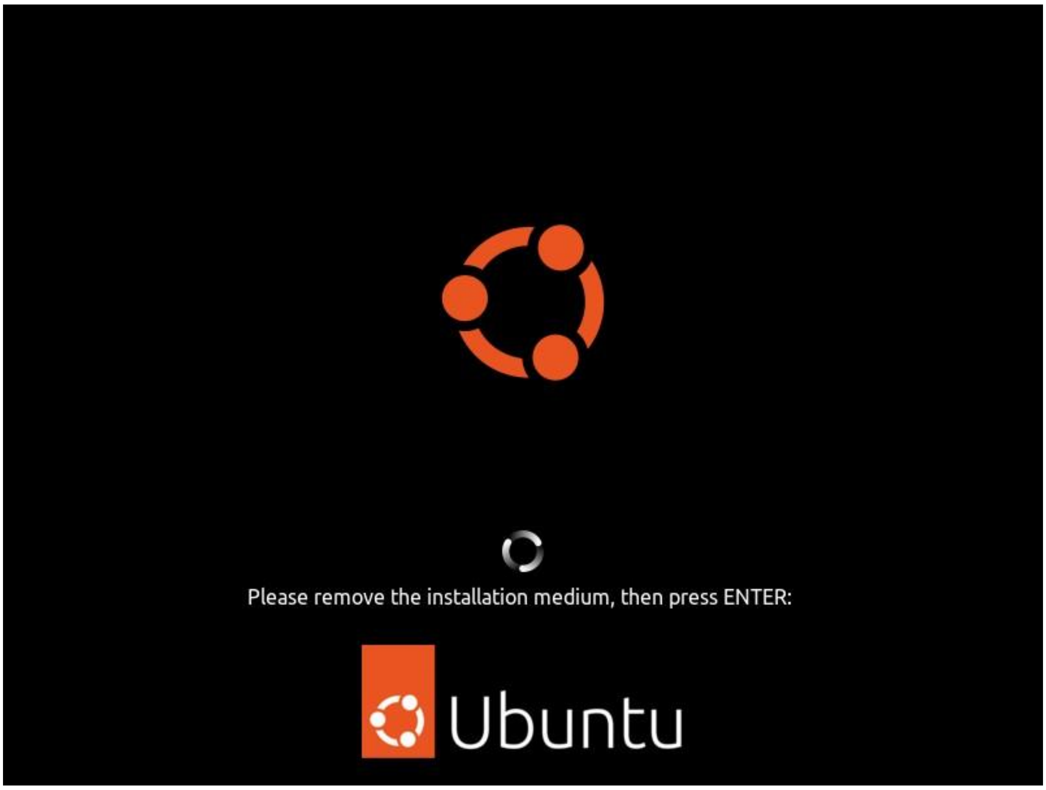
- Fill in Your Personal Details With Passwords and Click OnContinue.

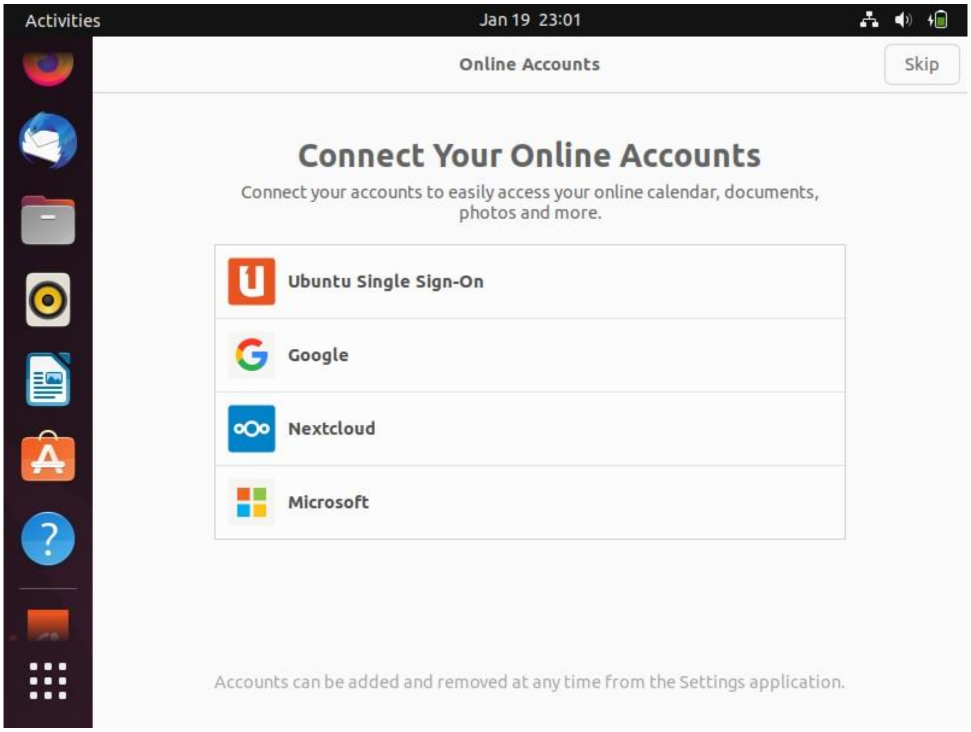




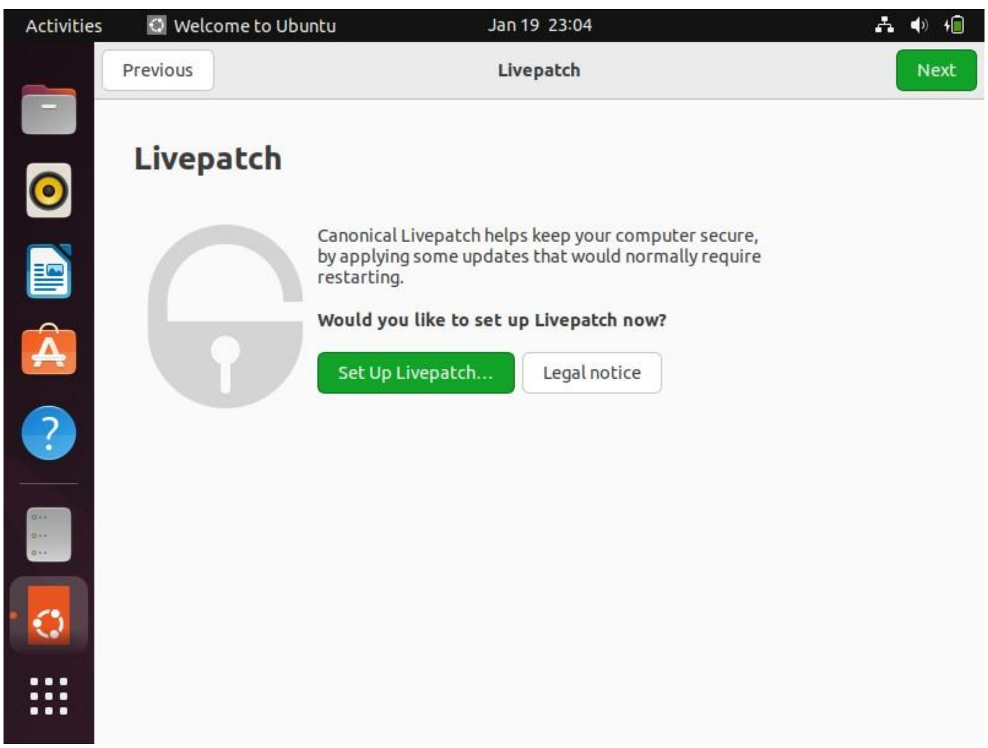


- Click On Restart Now

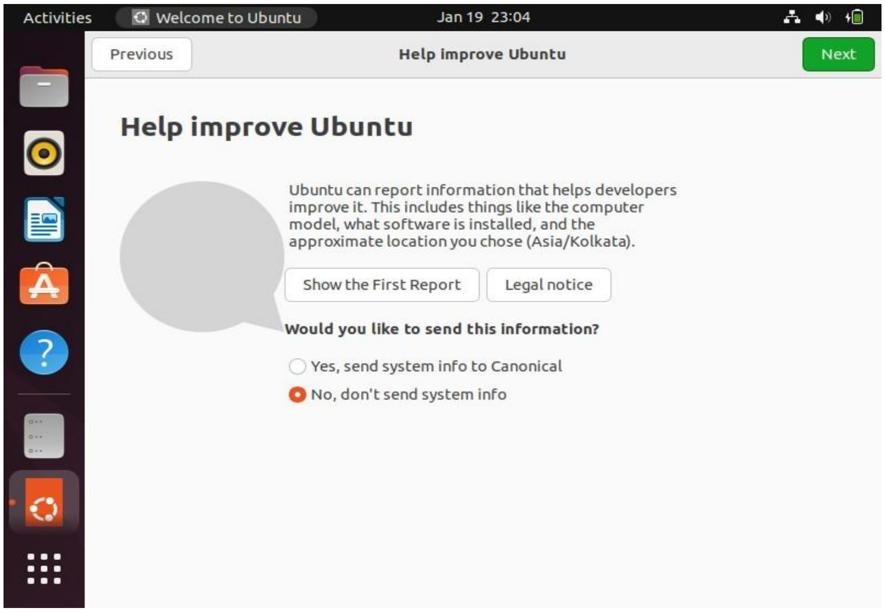




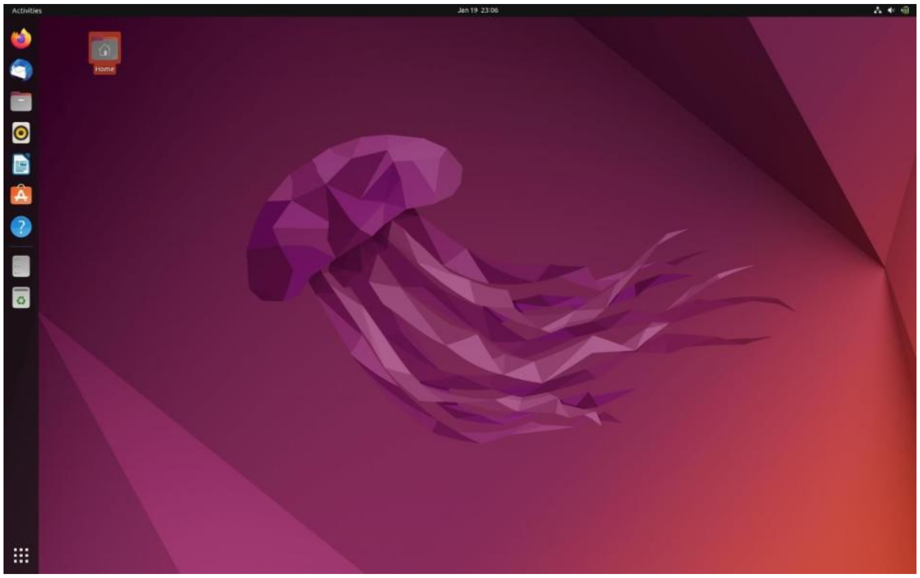
Setup Your Account



- Click On Next.



- Select Option "No, Don't Send System info then click on Next.



- Ubuntu Installation Is Completed.

Conclusion

The installation of the Linux/UNIX operating system has been completed successfully. The students gained knowledge of operating system setup, configuration, and management and gained hands-on experience in partitioning, bootloader configuration, and basic system customization.

References –

i. Textbook –

i. " Linux Administration: A Beginner's Guide", Wale Soyinka, Edition: 8th Edition (2020), Publisher: Addison-Wesley

ii." UNIX and Linux System Administration Handboo"- Kay A. Robbins, Edition: 5th Edition (2017),Publisher: Addison-Wesley Professional

ii. Online references –

a) https://www.amazon.com/Linux-Administration-Beginners-Guide-Eighth/dp/1260441714
b) https://www.amazon.com/Linux-Administration-Beginners-Guide-Eighth/dp/1260441714

Expected Oral Questions –

1. What are the steps involved in installing a Linux/UNIX operating system?
2. How do you create a bootable USB for installing Linux/UNIX?
3. What are the basic configurations required after installing a Linux/UNIX operating system?
4. What is the role of the GRUB bootloader in a Linux system?
5. What is the purpose of the root partition, home partition, and swap space?
6. What is dual booting, and how do you configure a system to dual boot Linux with another operating system?
7. What is a virtual machine, and how can Linux/UNIX be installed in a virtualized environment?
8. What steps would you take if the installation process fails or hangs?
9. How would you fix bootloader-related issues in Linux?